

Multimodal Retrieval-Augmented Generation (RAG) QA System

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Abstract. This report presents the design and implementation of a Multimodal Retrieval-Augmented Generation (RAG) Question Answering (QA) system. The system integrates text and image modalities to extract relevant information from PDF documents and provide precise answers. Using LangChain, FAISS, EasyOCR, and Transformer-based language models, our system offers robust document understanding, multimodal retrieval, and interactive querying via a Streamlit interface.

1 Introduction

Recent advances in Retrieval-Augmented Generation (RAG) models have significantly improved open-domain QA systems. However, most systems focus solely on textual data. This project introduces a multimodal RAG architecture capable of extracting and querying both text and image content (via OCR) from academic documents.

2 System Architecture

The system comprises the following components:

- **PDF Parser and Preprocessor:** Extracts text and images from uploaded PDFs.
- **Image OCR:** Uses EasyOCR to extract textual content from images.
- **Embedding and Vector Store:** Converts content into dense vectors using sentence-transformers and stores them using FAISS.
- **Retriever and Generator:** LangChain integrates retriever and HuggingFace generator to respond to queries.
- **Evaluation Module:** Computes BLEU, ROUGE, Cosine Similarity, and Response Time.
- **User Interface:** A Streamlit app enabling user-friendly interaction.

3 Methods

3.1 Text Extraction

Text is extracted from PDF using PyMuPDF and saved into a corpus.

3.2 Image OCR

Images are extracted using PyMuPDF and passed to EasyOCR to retrieve embedded textual data, which is combined with the main text corpus.

3.3 Embedding and Indexing

Both text and OCR-extracted image content are embedded using SentenceTransformers and stored in a FAISS vector store for efficient similarity-based retrieval.

3.4 Query Processing

When a user asks a question or uploads an image, embeddings are generated and matched against the FAISS index. Retrieved chunks are passed to a transformer-based LLM for final answer generation.

4 Evaluation

4.1 Metrics

We used the following metrics:

- **BLEU:** Measures n-gram overlap with a reference answer.
- **ROUGE-L:** Measures recall-oriented similarity.
- **Cosine Similarity:** Measures embedding similarity.
- **Response Time:** Measures system latency.

4.2 Results

The system was evaluated on a custom academic PDF. Table 1 shows average scores:

Table 1. Evaluation Results

Metric	Score
BLEU	0.78
ROUGE-L	0.81
Cosine Similarity	0.87
Response Time	1.23 sec

5 Challenges and Future Work

- **Image Quality:** OCR accuracy suffers from low-quality scanned images.
- **Multilingual PDFs:** OCR and embedding pipelines must handle diverse languages.
- **Scalability:** Currently limited to single PDF; future versions should support multi-document corpora and PDF classification.

6 Conclusion

Our system effectively demonstrates a scalable pipeline for building a multimodal RAG QA system. It integrates textual and visual content from documents, applies effective retrieval, and provides accurate answers with performance metrics. This lays a foundation for enterprise-grade academic assistants.

References

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